Bit-Parallel Finite-Domain Relational Search

Domain Unification as Bitwise AND

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Abstract

We show that finite-domain unification reduces to bitwise AND. When variable domains are represented as bitmasks, the unification goal $(x = y)$ computes the intersection $D_x \wedge D_y$ in a single CPU cycle. This enables 3,000–4,000× speedup over heap-based constraint operations via SIMD parallelism.

We implement this in Rust with AVX2/AVX-512 intrinsics and compare against: (1) stream-based miniKanren (reference semantics), (2) OCanren/faster-miniKanren (optimized miniKanren), (3) SWI-Prolog CLP(FD) (constraint propagation baseline).

The bit-parallel approach is complementary to full miniKanren: we accelerate the constraint propagation kernel while preserving relational semantics via a reference implementation.

1 Introduction

Scope Clarification. This paper addresses *Finite-Domain miniKanren* (FD-miniKanren), where variable domains are finite bitmasks. Standard miniKanren unification over unbounded term structures uses a separate reference implementation; the bit-parallel approach accelerates the constraint propagation core, not full structural unification.

2 The Core Insight: Domain Unification as AND

3 Implementation

4 Evaluation

4.1 Comparison with CLP(FD)

5 Related Work

6 Conclusion

The observation that finite-domain unification is bitwise AND enables parallelism while preserving the relational programming model.

Code Availability. `https://github.com/loveJesus/minikanren_1bit_chirho`

Companion Papers. This paper is part of a suite:

- Paper B: Relations as Sparse Boolean Tensors
- Paper C: Bridging Infinite Domains with Hash-Consed Term IDs
- Paper D: Tabling and Mode-Driven Goal Scheduling
- Paper E: Fully Fused Logic Accelerator (FPGA)
- Paper F: Differentiable Constraint Solving

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Soli Deo Gloria $\chi\rho$